Namespace LlamaLogic.TuningDescriptions

Classes

EnumTunableDescription

Represents the description of a tunable whose value is a member of an enumeration (a `TunableEnum` element) (🔒)

FragmentReferenceTunableDescription

Represents a reference to a shared description (a `TdescFragTag` element), whose Class names a TuningDescriptionFragment (🔒)

ListTunableDescription

Represents the description of a tunable that is a sequence of values described by Item (a `TunableList` element), including sets and mappings (🔒)

ScalarTunableDescription

Represents the description of a single-valued tunable (a `Tunable` element) (🔒)

SimDataGenerationException

Represents an exception thrown by SimDataGenerator when it cannot generate SimData for tuning

SimDataGenerator

Generates SimData resources from tuning the way Maxis' build does, using the tuning descriptions of the game version for which they are generated (🔒)

TunableDescription

Represents the description of a tunable (🔒)

TuningDescriptionAction

Represents a change Maxis recorded in a document's version history, identifying tunables by their Muid (🔒)

TuningDescriptionClass

Represents a class described in a TuningDescriptionModule (a `Class` element): an enumeration, the holder of module tuning, or both (🔒)

TuningDescriptionDocument

Represents a document of a TuningDescriptionSet: an instance, a module, or a fragment (🔒)

TuningDescriptionEnumItem

Represents a member of an enumeration described by a TuningDescriptionClass (an `EnumItem` element) (🔒)

TuningDescriptionFragment
: Represents a description shared by the FragmentReferenceTunableDescriptions that name it (a `TdescFrag` element), such as `TunableTestSet` (🔒)

TuningDescriptionInstance
: Represents the description of a class of instance tuning (an `Instance` element), such as `Buff` (🔒)

TuningDescriptionModule
: Represents the description of a Python module's tuning and enumerations (a `Module` element), such as `sims.sim_info_types` (🔒)

TuningDescriptionSet
: Represents a set of tuning descriptions for a version of The Sims 4, as Maxis distributes it (🔒)

TupleTunableDescription
: Represents the description of a tunable composed of named tunables (a `TunableTuple` element) (🔒)

UnknownTunableDescription
: Represents a description in an element this library does not recognize, kept so that nothing in a document is lost (🔒)

VariantTunableDescription
: Represents the description of a tunable whose value is one of several named options (a `TunableVariant` element) (🔒)

Enums

TuningDescriptionActionType
: The type of a TuningDescriptionAction

TuningDescriptionExportModes
: The destinations to which the game's build exports a tunable (its `export_modes` attribute)

Class EnumTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable whose value is a member of an enumeration (a `TunableEnum` element) (🔒)

```
public sealed class EnumTunableDescription : TunableDescription
```

Inheritance

object ← TunableDescription ← EnumTunableDescription

Inherited Members

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

BinaryType

Gets the type the tunable's value takes when it is exported (its `binary_type` attribute)

```
public string? BinaryType { get; }
```

Property Value

string

DynamicEntries

Gets the module tuning that adds members to the enumeration at run time (its `dynamic_entries` attribute), such as `tag.Tag._elements`

```
public string? DynamicEntries { get; }
```

Property Value

string

Remarks

Tuning descriptions do not contain these members; the game's module tuning does.

InvalidEntries

Gets the members of the enumeration tuning may not use (its `invalid_entries` attribute), as written

```
public string? InvalidEntries { get; }
```

Property Value

string

StaticEntries

Gets the enumeration the tunable's values come from (its `static_entries` attribute), which Get Enum(string) finds

```
public string? StaticEntries { get; }
```

Property Value

string

Type

Gets the name of the enumeration (its `type` attribute)

```
public string? Type { get; }
```

Property Value

string

Class FragmentReferenceTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a reference to a shared description (a `TdescFragTag` element), whose Class names a TuningDescriptionFragment (🔒)

```
public sealed class FragmentReferenceTunableDescription : TunableDescription
```

Inheritance

object ← TunableDescription ← FragmentReferenceTunableDescription

Inherited Members

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Resolve(TunableDescription) finds the description the reference stands for.

Class ListTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable that is a sequence of values described by Item (a `TunableList` element), including sets and mappings (🔒)

```
public sealed class ListTunableDescription : TunableDescription
```

Inheritance

object ← TunableDescription ← ListTunableDescription

Inherited Members

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

AreEntriesUnique

Gets whether tuning may not repeat an entry (its `unique_entries` attribute)

```
public bool AreEntriesUnique { get; }
```

Property Value

bool

IsMapping

Gets whether the list is a mapping, the items of which are pairs of the members named by MappingKey and MappingValue

```
public bool IsMapping { get; }
```

Property Value

bool

Item

Gets the description of the list's items, or null if the description has none

```
public TunableDescription? Item { get; }
```

Property Value

TunableDescription

MappingClass

Gets the name of the class of the mapping's pairs (its `mapping_class` attribute)

```
public string? MappingClass { get; }
```

Property Value

string

MappingKey

Gets the name of the member of Item that is the key of each pair (its `mapping_key` attribute)

```
public string? MappingKey { get; }
```

Property Value

string

MappingValue

Gets the name of the member of Item that is the value of each pair (its `mapping_value` attribute)

```
public string? MappingValue { get; }
```

Property Value

string

Class ScalarTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a single-valued tunable (a `Tunable` element) (🔒)

```
public sealed class ScalarTunableDescription : TunableDescription
```

Inheritance

object ← TunableDescription ← ScalarTunableDescription

Inherited Members

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

AllowNone

Gets whether the tunable may be `None` (its `allow_none` attribute), or null if the description does not say

```
public bool? AllowNone { get; }
```

Property Value

bool?

BinaryType

Gets the type the tunable's value takes when it is exported (its `binary_type` attribute)

```
public string? BinaryType { get; }
```

Property Value

string

Type

Gets the type of the tunable's value (its `type` attribute): a Python type such as `int`, `ResourceKey`, or the instance type of the tuning it references

```
public string? Type { get; }
```

Property Value

string

Class SimDataGenerationException

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents an exception thrown by SimDataGenerator when it cannot generate SimData for tuning

```
public sealed class SimDataGenerationException : Exception, ISerializable
```

Inheritance

object ← Exception ← SimDataGenerationException

Implements

ISerializable

Inherited Members

Exception.GetBaseException() , Exception.GetObjectData(SerializationInfo, StreamingContext) , Exception.GetType() , Exception.ToString() , Exception.Data , Exception.HelpLink , Exception.HResult , Exception.InnerException , Exception.Message , Exception.Source , Exception.StackTrace , Exception.TargetSite , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.ReferenceEquals(object, object)

Constructors

SimDataGenerationException()

Initializes a new SimDataGenerationException

```
public SimDataGenerationException()
```

SimDataGenerationException(Exception?)

Initializes a new SimDataGenerationException with the specified `innerException`

```
public SimDataGenerationException(Exception? innerException)
```

Parameters

`innerException` Exception

SimDataGenerationException(string)

Initializes a new SimDataGenerationException with the specified `message`

```
public SimDataGenerationException(string message)
```

Parameters

`message` string

SimDataGenerationException(string, Exception?)

Initializes a new SimDataGenerationException with the specified `message` and `innerException`

```
public SimDataGenerationException(string message, Exception? innerException)
```

Parameters

`message` string

`innerException` Exception

Class SimDataGenerator

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Generates SimData resources from tuning the way Maxis' build does, using the tuning descriptions of the game version for which they are generated (🔒)

```
public sealed class SimDataGenerator
```

Inheritance

object ← SimDataGenerator

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Which tunables become columns, the columns' types, and the values written for tuning that omits a tunable all come from Descriptions.

Some of what Maxis' build knows is not in tuning descriptions, so the generator learns it:

- AddTuning(XDocument) and AddTuning(CombinedTuningModel) learn the entries of dynamic enumerations (such as tags defined by module tuning) and the instances of tuning that is referenced by name
- AddLayouts(SimDataModel) learns the column order of schemas the game reads with a layout of its own (such as `Client_*` and `Renderer.*` schemas), whose columns are otherwise ordered by name

Maxis' build also converts some resource keys: a PNGImage key in tuning is written to SimData as a DSTImage key, and an AudioConfigurationReference key as an AudioConfiguration key in group `0x001407EC`. The generator does the same unless RemoveResourceKeyConversion(ResourceType, uint) says otherwise, and AddResourceKeyConversion(ResourceType, uint, ResourceType, uint) adds conversions.

For the game's own SimData, the game's combined tuning and SimData are the sources of what the generator needs to learn.

Constructors

SimDataGenerator(TuningDescriptionSet)

Initializes a new SimDataGenerator

```
public SimDataGenerator(TuningDescriptionSet descriptions)
```

Parameters

`descriptions` TuningDescriptionSet

The tuning descriptions of the game version for which SimData will be generated

Properties

Descriptions

Gets the tuning descriptions of the game version for which SimData is generated

```
public TuningDescriptionSet Descriptions { get; }
```

Property Value

TuningDescriptionSet

Methods

AddLayouts(SimDataModel)

Learns the column order of each schema in the specified `simData` whose columns are not ordered by name

```
public void AddLayouts(SimDataModel simData)
```

Parameters

`simData` SimDataModel

Remarks

When generating an object of a schema with the same name and the same columns, the generator orders the columns as they are ordered in `simData`.

AddResourceKeyConversion(ResourceType, uint, ResourceType, uint)

Learns that resource keys of the specified `fromType` and `fromGroup` in tuning are written to SimData with the specified `toType` and `toGroup`

```
public void AddResourceKeyConversion(ResourceType fromType, uint fromGroup, ResourceType toType, uint toGroup)
```

Parameters

`fromType` ResourceType

`fromGroup` uint

`toType` ResourceType

`toGroup` uint

AddTuning(CombinedTuningModel)

Learns the entries of dynamic enumerations and the names of instances from each tuning in the specified `combinedTuning`

```
public void AddTuning(CombinedTuningModel combinedTuning)
```

Parameters

`combinedTuning` CombinedTuningModel

AddTuning(XDocument)

Learns the entries of dynamic enumerations the specified `tuning` defines (if it is module tuning) and the name of its instance

```
public void AddTuning(XDocument tuning)
```

Parameters

`tuning` XDocument

Remarks

An instance or enumeration value that is not a decimal number (such as one a modding tool wrote in hexadecimal) is not learned.

Generate(XDocument)

Generates the SimData for the specified `tuning`

```
public SimDataModel Generate(XDocument tuning)
```

Parameters

`tuning` XDocument

Instance tuning (an `I` element) or module tuning (an `M` element)

Returns

SimDataModel

The SimData, with one instance for instance tuning or one instance for each class of module tuning that has exported tunables

Remarks

The Pack of the SimData is `0`, the base game's; set it for SimData that belongs to a pack.

Exceptions

SimDataGenerationException

`tuning` cannot be generated with Descriptions and what the generator has learned

IsWrittenToSimData(TuningDescriptionInstance, IReadOnlyList<TunableDescription>)

Gets whether the tunable at the end of the specified `path` in instance tuning of the specified `instance` class is written to the SimData Generate(XDocument) generates, so that changing it changes the SimData

```
public bool IsWrittenToSimData(TuningDescriptionInstance instance,
IReadOnlyList<TunableDescription> path)
```

Parameters

`instance` TuningDescriptionInstance

The class of the instance tuning

`path` IReadOnlyList <TunableDescription>

The descriptions from one of the class's effective members (see GetEffectiveMembers()) down to the tunable: after a list, its item; after a variant, the option; after a tuple, the member

Returns

bool

Remarks

The same rules decide this as decide what Generate(XDocument) writes, including those that export a whole class (its `export_modes` has `client_binary`), a member through the class that declares it, and every member of a tuple none of whose members is exported. A path that ends at a variant's option stands for choosing that option, which changes the variant's hash, so it is written whenever the variant is, even when the option's own tunables are not.

RemoveResourceKeyConversion(ResourceType, uint)

Stops converting resource keys of the specified `fromType` and `fromGroup` in tuning, so that they are written to SimData as they are

```
public void RemoveResourceKeyConversion(ResourceType fromType, uint fromGroup)
```

Parameters

`fromType` ResourceType

`fromGroup` uint

Class TunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable (🔒)

```
public abstract class TunableDescription
```

Inheritance

object ← TunableDescription

Derived

EnumTunableDescription, FragmentReferenceTunableDescription, ListTunableDescription, ScalarTunableDescription, TupleTunableDescription, UnknownTunableDescription, VariantTunableDescription

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Every attribute of the description's XML element is available in Attributes, including those this library does not give properties of their own.

Properties

Attributes

Gets all of the description's attributes, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

Property Value

IReadOnlyDictionary<string, string>

Class

Gets the Python class of the tunable (its `class` attribute)

```
public string? Class { get; }
```

Property Value

string

Default

Gets the tunable's default value, as written (its `default` attribute)

```
public string? Default { get; }
```

Property Value

string

Description

Gets the description of the tunable for tuners (its `description` attribute)

```
public string? Description { get; }
```

Property Value

string

Display

Gets the display name of the tunable (its `display` attribute)

```
public string? Display { get; }
```

Property Value

string

ExportModes

Gets the destinations to which the tunable is exported (its `export_modes` attribute), or null if the description does not specify them

```
public TuningDescriptionExportModes? ExportModes { get; }
```

Property Value

TuningDescriptionExportModes?

Group

Gets the group of the tunable in tuning editors (its `group` attribute)

```
public string? Group { get; }
```

Property Value

string

IsDeprecated

Gets whether the tunable is deprecated (its `Deprecated` attribute)

```
public bool IsDeprecated { get; }
```

Property Value

bool

Members

Gets the descriptions nested in this one, in document order

```
public IReadOnlyList<TunableDescription> Members { get; }
```

Property Value

IReadOnlyList<TunableDescription>

Muid

Gets the unique identifier Maxis assigns the tunable (its `muid` attribute)

```
public ulong? Muid { get; }
```

Property Value

ulong?

Name

Gets the name of the tunable (its `name` attribute), or null for the item of a list

```
public string? Name { get; }
```

Property Value

string

Methods

GetMember(string)

Gets the nested description with the specified `name`, or null if there is none

```
public TunableDescription? GetMember(string name)
```

Parameters

`name` string

Returns

TunableDescription

Class TuningDescriptionAction

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a change Maxis recorded in a document's version history, identifying tunables by their Muid (🔒)

```
public sealed class TuningDescriptionAction
```

Inheritance

object ← TuningDescriptionAction

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Attributes

Gets all of the action's attributes, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

Property Value

IReadOnlyDictionary<string, string>

NewMuid

Gets the MUID that replaces OldMuid (its `new_muid` attribute)

```
public ulong? NewMuid { get; }
```

Property Value

ulong?

OldMuid

Gets the MUID the action removes or renames (its `old_muid` attribute)

```
public ulong? OldMuid { get; }
```

Property Value

ulong?

SourceTypeValue

Gets the option value the action changes from (its `source_type_value` attribute)

```
public string? SourceTypeValue { get; }
```

Property Value

string

TargetMuid

Gets the MUID of the tunable the action changes (its `target_muid` attribute)

```
public ulong? TargetMuid { get; }
```

Property Value

ulong?

TargetTypeValue

Gets the option value the action changes to (its `target_type_value` attribute)

```
public string? TargetTypeValue { get; }
```

Property Value

string

Type

Gets the type of the action

```
public TuningDescriptionActionType Type { get; }
```

Property Value

TuningDescriptionActionType

TypeName

Gets the type of the action as written (its `type` attribute)

```
public string TypeName { get; }
```

Property Value

string

Enum TuningDescriptionActionType

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

The type of a TuningDescriptionAction

```
public enum TuningDescriptionActionType
```

Fields

`ChangeInheritance = 4`

A class's inheritance changed (`kChangeInheritance`)

`Remove = 1`

A tunable was removed (`kRemove`)

`Rename = 2`

A tunable was renamed (`kRename`)

`Unknown = 0`

A type this library does not recognize (see TypeName)

`UpdateVariantTypeField = 3`

The value naming a variant's option changed (`kUpdateVariantTypeField`)

Class TuningDescriptionClass

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a class described in a TuningDescriptionModule (a `Class` element): an enumeration, the holder of module tuning, or both (🔒)

```
public sealed class TuningDescriptionClass
```

Inheritance

object ← TuningDescriptionClass

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Attributes

Gets all of the class's attributes, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

Property Value

IReadOnlyDictionary<string, string>

Classes

Gets the classes nested in this one, in document order

```
public IReadOnlyList<TuningDescriptionClass> Classes { get; }
```

Property Value

IReadOnlyList<TuningDescriptionClass>

EnumItems

Gets the members of the enumeration the class is, in document order

```
public IReadOnlyList<TuningDescriptionEnumItem> EnumItems { get; }
```

Property Value

IReadOnlyList<TuningDescriptionEnumItem>

IsBitFlagEnum

Gets whether the class is an enumeration of bit flags (its `enum_bit_flag` attribute)

```
public bool IsBitFlagEnum { get; }
```

Property Value

bool

IsEnum

Gets whether the class is an enumeration

```
public bool IsEnum { get; }
```

Property Value

bool

IsLockedEnum

Gets whether the enumeration's members are fixed (its `enum_locked` attribute)

```
public bool IsLockedEnum { get; }
```

Property Value

bool

Members

Gets the module tuning the class declares, in document order

```
public IReadOnlyList<TunableDescription> Members { get; }
```

Property Value

IReadOnlyList<TunableDescription>

Module

Gets the module in which the class is described

```
public TuningDescriptionModule Module { get; }
```

Property Value

TuningDescriptionModule

Muid

Gets the unique identifier Maxis assigns the class (its `muid` attribute)

```
public ulong? Muid { get; }
```

Property Value

ulong?

Name

Gets the name of the class (its `name` attribute)

```
public string Name { get; }
```

Property Value

string

Parent

Gets the class in which this one is nested, or null if it is not nested

```
public TuningDescriptionClass? Parent { get; }
```

Property Value

TuningDescriptionClass

StaticEntriesName

Gets the name by which a StaticEntries attribute refers to the class, such as `sims-sim_info_types.Age`

```
public string StaticEntriesName { get; }
```

Property Value

string

Methods

GetClass(string)

Gets the nested class with the specified `name`, or null if there is none

```
public TuningDescriptionClass? GetClass(string name)
```

Parameters

`name` string

Returns

TuningDescriptionClass

GetEnumValue(string)

Gets the value of the enumeration member with the specified `name`, or null if there is none

```
public long? GetEnumValue(string name)
```

Parameters

`name` string

Returns

long?

Class TuningDescriptionDocument

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a document of a TuningDescriptionSet: an instance, a module, or a fragment (🔒)

```
public abstract class TuningDescriptionDocument
```

Inheritance

object ← TuningDescriptionDocument

Derived

TuningDescriptionFragment, TuningDescriptionInstance, TuningDescriptionModule

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.MemberwiseClone() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Attributes

Gets all of the attributes of the document's root element, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

Property Value

IReadOnlyDictionary<string, string>

Muid

Gets the unique identifier Maxis assigns the document (its `muid` attribute)

```
public ulong? Muid { get; }
```

Property Value

ulong?

Path

Gets the path of the document in its set, with `/` separating directories

```
public string Path { get; }
```

Property Value

string

Set

Gets the set to which the document belongs

```
public TuningDescriptionSet Set { get; }
```

Property Value

TuningDescriptionSet

VersionHistory

Gets the changes Maxis recorded for the document, one list per `ActionList` element, in document order

```
public IReadOnlyList<IReadOnlyList<TuningDescriptionAction>> VersionHistory { get; }
```

Property Value

IReadOnlyList<IReadOnlyList<TuningDescriptionAction>>

Class TuningDescriptionEnumItem

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a member of an enumeration described by a TuningDescriptionClass (an `EnumItem` element) (🔒)

```
public sealed class TuningDescriptionEnumItem
```

Inheritance

object ← TuningDescriptionEnumItem

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Attributes

Gets all of the member's attributes, by name

```
public IReadOnlyDictionary<string, string> Attributes { get; }
```

Property Value

IReadOnlyDictionary<string, string>

Muid

Gets the unique identifier Maxis assigns the member (its `muid` attribute)

```
public ulong? Muid { get; }
```

Property Value

ulong?

Name

Gets the name of the member (its `name` attribute)

```
public string Name { get; }
```

Property Value

string

Value

Gets the value of the member (its `enum_value` attribute)

```
public long Value { get; }
```

Property Value

long

Enum TuningDescriptionExportModes

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

The destinations to which the game's build exports a tunable (its `export_modes` attribute)

```
[Flags]
public enum TuningDescriptionExportModes
```

Fields

`ClientBinary = 1`

The tunable is exported to `SimData` resources (`client_binary`)

`None = 0`

No destinations

`ServerBinary = 2`

The tunable is exported to the binary combined tuning (`server_binary`)

`ServerXml = 4`

The tunable is exported to tuning XML (`server_xml`)

Class TuningDescriptionFragment

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a description shared by the FragmentReferenceTunableDescriptions that name it (a `TdescFrag` element), such as `TunableTestSet` (🔒)

```
public sealed class TuningDescriptionFragment : TuningDescriptionDocument
```

Inheritance

object ← TuningDescriptionDocument ← TuningDescriptionFragment

Inherited Members

TuningDescriptionDocument.Attributes , TuningDescriptionDocument.Muid , TuningDescriptionDocument.Path , TuningDescriptionDocument.Set , TuningDescriptionDocument.VersionHistory , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Class

Gets the name by which references refer to the fragment (its description's `class` attribute)

```
public string Class { get; }
```

Property Value

string

Description

Gets the description the fragment shares

```
public TunableDescription Description { get; }
```

Property Value

TunableDescription

Class TuningDescriptionInstance

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a class of instance tuning (an `Instance` element), such as `Buff` (🔒)

```
public sealed class TuningDescriptionInstance : TuningDescriptionDocument
```

Inheritance

object ← TuningDescriptionDocument ← TuningDescriptionInstance

Inherited Members

TuningDescriptionDocument.Attributes , TuningDescriptionDocument.Muid , TuningDescriptionDocument.Path , TuningDescriptionDocument.Set , TuningDescriptionDocument.VersionHistory , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Members holds only the tunables the class declares; GetEffectiveMembers() adds those it inherits.

Properties

Class

Gets the Python class of the instance tuning (its `class` attribute)

```
public string Class { get; }
```

Property Value

string

DeletedMembers

Gets the names of the inherited tunables the class removes (its `Deleted` elements)

```
public IReadOnlyList<string> DeletedMembers { get; }
```

Property Value

IReadOnlyList <string>

Description

Gets the description of the class for tuners (its `description` attribute)

```
public string? Description { get; }
```

Property Value

string

ExportModes

Gets the destinations to which the whole class is exported (its `export_modes` attribute), or null if the description does not specify them

```
public TuningDescriptionExportModes? ExportModes { get; }
```

Property Value

TuningDescriptionExportModes?

InstanceType

Gets the type of instance tuning the class is (its `instance_type` attribute), such as `buff`

```
public string? InstanceType { get; }
```

Property Value

string

Members

Gets the tunables the class declares, in document order

```
public IReadOnlyList<TunableDescription> Members { get; }
```

Property Value

IReadOnlyList<TunableDescription>

Module

Gets the Python module that defines the class (its `module` attribute)

```
public string? Module { get; }
```

Property Value

string

Parents

Gets the classes from which the class inherits (its `parents` attribute), in declaration order

```
public IReadOnlyList<string> Parents { get; }
```

Property Value

IReadOnlyList<string>

Methods

GetEffectiveMembers()

Gets the tunables of the class, including those it inherits and excluding those it deletes

```
public IReadOnlyList<TunableDescription> GetEffectiveMembers()
```

Returns

IReadOnlyList<TunableDescription>

Remarks

Each tunable is the description from the first class in GetMethodResolutionOrder() that declares or deletes it, as in Python. The tunables are in the order the game collects them: walking from the most basic class to this one, each class first removes the tunables it deletes and then adds or replaces those it declares, a newly added tunable going to the end.

Exceptions

InvalidDataException

The class's inheritance cannot be resolved (see GetMethodResolutionOrder())

GetMember(string)

Gets the tunable the class declares with the specified `name`, or null if it declares none

```
public TunableDescription? GetMember(string name)
```

Parameters

`name` string

Returns

TunableDescription

GetMethodResolutionOrder()

Gets this class followed by the classes from which it inherits, in the order Python searches them (the C3 linearization of Parents)

```
public IReadOnlyList<TuningDescriptionInstance> GetMethodResolutionOrder()
```

Returns

IReadOnlyList <TuningDescriptionInstance>

Exceptions

InvalidDataException

A parent is not in the set, the inheritance is circular, or the parents cannot be linearized

Class TuningDescriptionModule

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a Python module's tuning and enumerations (a `Module` element), such as `sims.sim_info_types` (🔒)

```
public sealed class TuningDescriptionModule : TuningDescriptionDocument
```

Inheritance

object ← TuningDescriptionDocument ← TuningDescriptionModule

Inherited Members

TuningDescriptionDocument.Attributes , TuningDescriptionDocument.Muid , TuningDescriptionDocument.Path , TuningDescriptionDocument.Set , TuningDescriptionDocument.VersionHistory , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

Classes

Gets the module's classes, in document order

```
public IReadOnlyList<TuningDescriptionClass> Classes { get; }
```

Property Value

IReadOnlyList<TuningDescriptionClass>

Name

Gets the name of the module (its `name` attribute)

```
public string Name { get; }
```

Property Value

string

Methods

GetClass(string)

Gets the class of the module with the specified `name`, or null if there is none

```
public TuningDescriptionClass? GetClass(string name)
```

Parameters

`name` string

Returns

TuningDescriptionClass

Class TuningDescriptionSet

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a set of tuning descriptions for a version of The Sims 4, as Maxis distributes it (🔒)

```
public sealed class TuningDescriptionSet : IDisposable
```

Inheritance

object ← TuningDescriptionSet

Implements

IDisposable

Inherited Members

object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

A set is indexed by the paths of its documents when it is opened: instances are `*/Descriptions/{class}.tdesc`, modules are `Tuning/{module with . replaced by -}.tdesc`, and fragments are `TdescFrag/{class}.tdescfrag`. A document is read the first time it is requested, and it may be read again once nothing refers to it any longer, so requesting a document twice does not necessarily return the same object.

A set's members are safe to use from multiple threads.

Properties

FragmentClasses

Gets the names of the classes of the set's fragments

```
public IReadOnlyCollection<string> FragmentClasses { get; }
```

Property Value

IReadOnlyCollection<string>

GameVersion

Gets the version of the game the set describes, if it was opened from a file named as Maxis names them

```
public Version? GameVersion { get; }
```

Property Value

Version

InstanceClasses

Gets the names of the classes of the set's instances

```
public IReadOnlyCollection<string> InstanceClasses { get; }
```

Property Value

IReadOnlyCollection<string>

InstanceTypes

Gets the types of instance tuning the set's instances are (their `instance_type` attributes), such as `buff`

```
public IReadOnlyCollection<string> InstanceTypes { get; }
```

Property Value

IReadOnlyCollection<string>

Remarks

The types are read from the beginning of each instance's document, which is much faster than reading every instance.

Exceptions

InvalidDataException

An instance's document is malformed

ModuleNames

Gets the names of the set's modules

```
public IReadOnlyCollection<string> ModuleNames { get; }
```

Property Value

IReadOnlyCollection<string>

Methods

Dispose()

Performs application-defined tasks associated with freeing, releasing, or resetting unmanaged resources.

```
public void Dispose()
```

FromDirectory(string)

Opens the set of tuning descriptions extracted to the directory at the specified `path` ( )

```
public static TuningDescriptionSet FromDirectory(string path)
```

Parameters

`path` string

Returns

TuningDescriptionSet

Exceptions

DirectoryNotFoundException

There is no directory at `path`

FromZip(Stream)

Opens the ZIP archive of tuning descriptions in the specified `stream`, which must remain open and seekable until the set is disposed (🗑)

```
public static TuningDescriptionSet FromZip(Stream stream)
```

Parameters

`stream` Stream

Returns

TuningDescriptionSet

FromZip(string)

Opens the ZIP archive of tuning descriptions at the specified `path` (🗑)

```
public static TuningDescriptionSet FromZip(string path)
```

Parameters

`path` string

Returns

TuningDescriptionSet

Remarks

When the name of the file is that of a Maxis distribution (such as `1.128.90.1030-TDESC.zip`), Game Version is the version it names.

GetEnum(string)

Gets the enumeration to which an StaticEntries attribute refers, such as `sims-sim_info_types.Age`, or null if the set does not describe it

```
public TuningDescriptionClass? GetEnum(string staticEntries)
```

Parameters

`staticEntries` string

Returns

TuningDescriptionClass

Exceptions

InvalidDataException

The module that would describe the enumeration is malformed

GetFragment(string)

Gets the fragment with the specified `className`, or null if the set does not have it

```
public TuningDescriptionFragment? GetFragment(string className)
```

Parameters

`className` string

Returns

TuningDescriptionFragment

Exceptions

InvalidDataException

The fragment's document is malformed or describes a different fragment

GetInstance(string)

Gets the instance with the specified `className`, or null if the set does not have it

```
public TuningDescriptionInstance? GetInstance(string className)
```

Parameters

`className` string

Returns

TuningDescriptionInstance

Exceptions

InvalidDataException

The instance's document is malformed or describes a different class

GetModule(string)

Gets the module with the specified `name`, such as `sims.sim_info_types`, or null if the set does not have it

```
public TuningDescriptionModule? GetModule(string name)
```

Parameters

`name` string

Returns

TuningDescriptionModule

Exceptions

InvalidDataException

The module's document is malformed or describes a different module

Resolve(TunableDescription)

Gets the description the specified `description` stands for: the description of the fragment it refers to if it is a FragmentReferenceTunableDescription, and otherwise itself

```
public TunableDescription Resolve(TunableDescription description)
```

Parameters

`description` TunableDescription

Returns

TunableDescription

Exceptions

InvalidDataException

The set does not have the fragment to which `description` refers

Class TupleTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable composed of named tunables (a `TunableTuple` element) (🔒)

```
public sealed class TupleTunableDescription : TunableDescription
```

Inheritance

object ← TunableDescription ← TupleTunableDescription

Inherited Members

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Class UnknownTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents a description in an element this library does not recognize, kept so that nothing in a document is lost (🔒)

```
public sealed class UnknownTunableDescription : TunableDescription
```

Inheritance

object ← TunableDescription ← UnknownTunableDescription

Inherited Members

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Properties

ElementName

Gets the name of the description's XML element

```
public string ElementName { get; }
```

Property Value

string

Class VariantTunableDescription

Namespace: LlamaLogic.TuningDescriptions

Assembly: LlamaLogic.TuningDescriptions.dll

Represents the description of a tunable whose value is one of several named options (a `TunableVariant` element) (🔒)

```
public sealed class VariantTunableDescription : TunableDescription
```

Inheritance

object ← TunableDescription ← VariantTunableDescription

Inherited Members

TunableDescription.Attributes , TunableDescription.Class , TunableDescription.Default , TunableDescription.Description , TunableDescription.Display , TunableDescription.ExportModes , TunableDescription.Group , TunableDescription.IsDeprecated , TunableDescription.Members , TunableDescription.Muid , TunableDescription.Name , TunableDescription.GetMember(string) , object.Equals(object) , object.Equals(object, object) , object.GetHashCode() , object.GetType() , object.ReferenceEquals(object, object) , object.ToString()

Remarks

Default is the name of the option tuning gets when it does not choose one.

Properties

Options

Gets the options of the variant, in document order

```
public IReadOnlyList<TunableDescription> Options { get; }
```

Property Value

IReadOnlyList<TunableDescription>

Methods

GetOption(string)

Gets the option with the specified `name`, or null if there is none

```
public TunableDescription? GetOption(string name)
```

Parameters

`name` string

Returns

TunableDescription